

Linear systems and elimination

Is the ideal GMRES bound sharp for every Jordan block?

Difficulty: challenging **Importance:** interesting to specialist

Status: open; several parameter regimes are known.

Problem statement

For $n \geq 2$ and $\lambda \in \mathbb{C} \setminus \{0\}$, let $J_n(\lambda) = \lambda I + N$, where $N_{i,i+1} = 1$ and all other entries of N vanish. Let $\mathcal{D}_k = \{p \in \mathbb{C}[z] : \deg p \leq k, p(0) = 1\}$. Define

$$\psi_k(J) = \max_{\|v\|_2=1} \min_{p \in \mathcal{D}_k} \|p(J)v\|_2, \quad \phi_k(J) = \min_{p \in \mathcal{D}_k} \|p(J)\|_2.$$

Prove or disprove $\psi_k(J_n(\lambda)) = \phi_k(J_n(\lambda))$ for every $1 \leq k < n$. The maximum describes the slowest possible GMRES residual reduction, whereas the minimum over operator norms is the ideal bound. The familiar inequality $\psi_k \leq \phi_k$ does not answer the question. A single Jordan block is a published structural test case, not a claim about all nonnormal matrices.

References

Tichý, Liesen, and Faber, *On worst-case GMRES, ideal GMRES, and the polynomial numerical hull of a Jordan block*, ETNA 26 (2007), 453–473, §1 conjecture and subsequent special cases. Faber, Liesen, and Tichý, *Matrix best approximation in the spectral norm*, published in LAA 733 (2026), §§4–5.

Status check

Searches for *Jordan block ideal GMRES equality proved 2026* and *site:arxiv.org GMRES Jordan block* found the original partial results and the 2026 general approximation paper, but no resolution of the displayed Jordan-block equality. The latter paper's doubling theorem changes the matrix and does not by itself establish this statement.